Homework #8

1) Given the C program program for serial communication. Answer the questions for parts a and b below.

```
#include <avr/io.h>

unsigned char cleared = 1<<UDRE;

int main()
{
        int i;
        DDRD = 0b00000010;
        PORTD = 0xFF;
        UCSRB = (1<<RXEN) | (1<<TXEN);
        UCSRC = (1<<UCSZ1) | (1<<UCSZ0) | (1<<URSEL);
        UBRRL = 0x33;
        while(1)
        {
                while(!(UCSRA & cleared)); // POLLING
                UDR = 'G';
        }

        return 0;
}
```

a. Is serial transmission or reception or both modes enabled? Which serial operation(s) is(are) performed in the program (transmission/reception/both)? Circle the program instruction(s) that you used to determine your answer.

Serial transmission and reception are enabled but only transmission is performed. See highlighted instructions in red.

b. Is polling or an interrupt used in this program for serial communication? Circle the program instruction(s) that you used to determine your answer.

See red highlighted instruction with POLLING as comment above.

2) Below is a C program that outputs square waves using Timers 0 and 2. Create a Microchip Studio workspace (GCC C Executable Project for the ATmega32) and copy the C code into the C workspace. Answer the questions for parts a-e on the following page.

```
#include <AVR/IO.H>
#include <AVR/INTERRUPT.H>

ISR(TIMER0_OVF_vect);
ISR(TIMER2_OVF_vect);

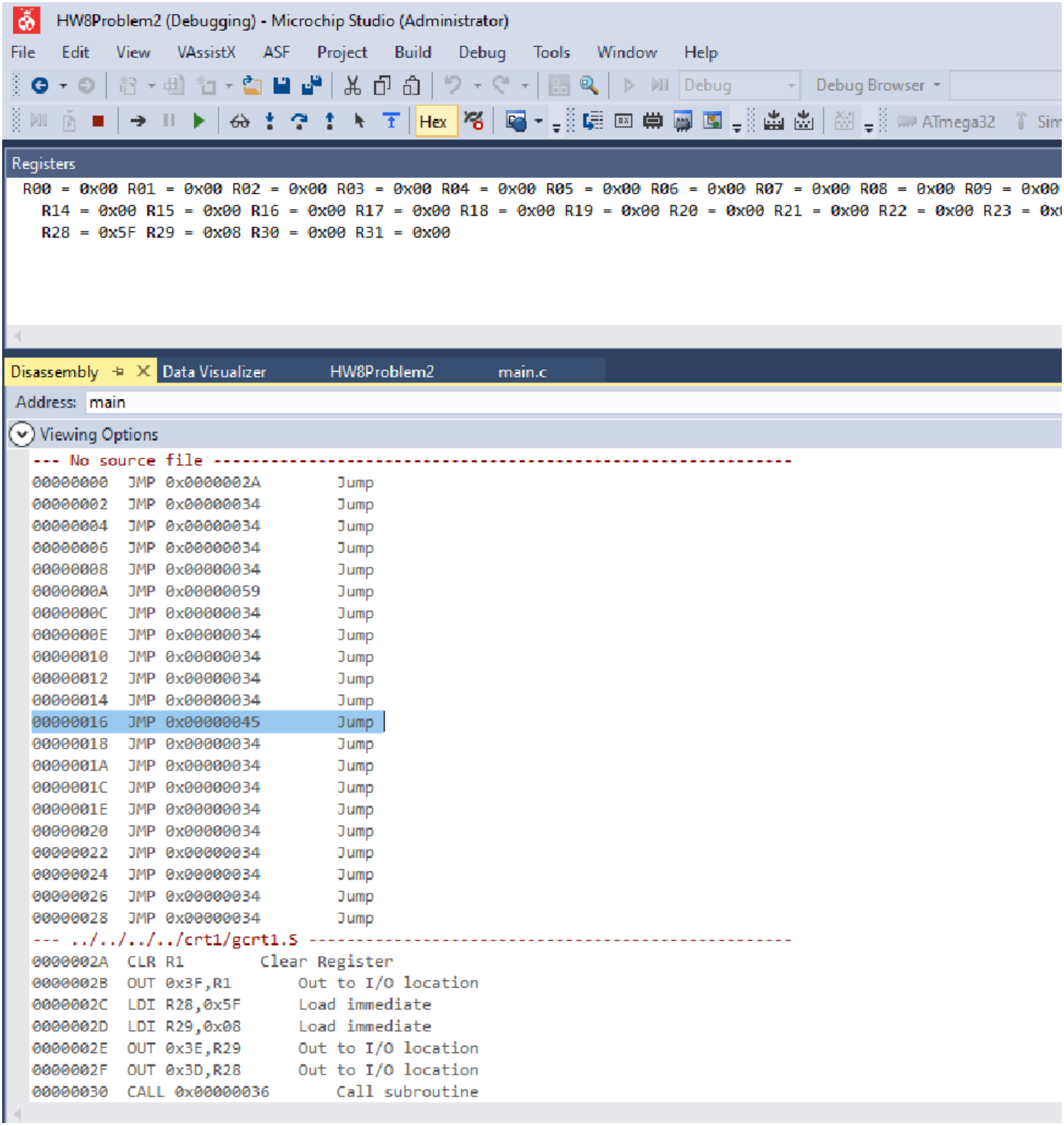

int main()
```

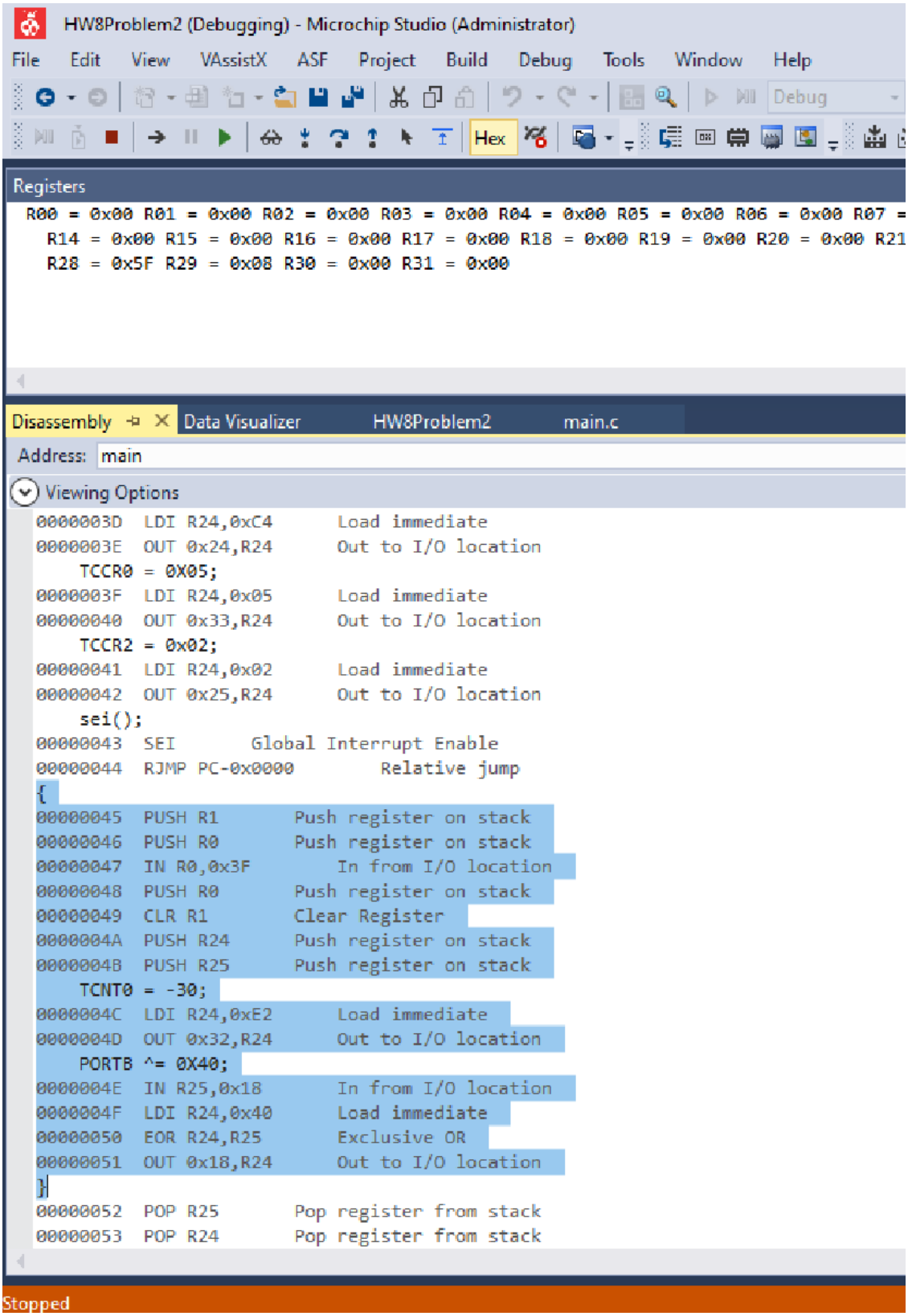
```
{
	DDRB |= 0B01100000;
	TIMSK = (1<<TOIE0) | (1<<TOIE2);
	TCNT0 = -30;
	TCNT2 = -60;
	TCCR0 = 0X05;
	TCCR2 = 0x02;
	sei();
	while(1);
	return 1;
}

ISR(TIMER0_OVF_vect)
{
	TCNT0 = -30;
	PORTB ^= 0X40;
}

ISR(TIMER2_OVF_vect)
{
	TCNT2 = -60;
	PORTB ^= 0X20;
}
```

a) From the Disassembly window of the workspace, identify the interrupt code for the interrupt routines for Timers 0 and 2. Submit a screenshot of showing the C/assembly code for both interrupt service routines.


HW8Problem2 (Debugging) - Microchip Studio (Administrator)
File Edit View VAssistX ASF Project Build Debug Tools Window Help
Debug
Debug Browser
Hex
ATmega32
Registers
R00 = 0x00 R01 = 0x00 R02 = 0x00 R03 = 0x00 R04 = 0x00 R05 = 0x00 R06 = 0x00 R07 = 0x00 R08 = 0x00 R09 = 0x00
R14 = 0x00 R15 = 0x00 R16 = 0x00 R17 = 0x00 R18 = 0x00 R19 = 0x00 R20 = 0x00 R21 = 0x00 R22 = 0x00 R23 = 0x
R28 = 0x5F R29 = 0x08 R30 = 0x00 R31 = 0x00
Disassembly
Data Visualizer
HW8Problem2
main.c
Address: main
Viewing Options
--- No source file ---
00000000 JMP 0x0000002A Jump
00000002 JMP 0x00000034 Jump
00000004 JMP 0x00000034 Jump
00000006 JMP 0x00000034 Jump
00000008 JMP 0x00000034 Jump
0000000A JMP 0x00000059 Jump
0000000C JMP 0x00000034 Jump
0000000E JMP 0x00000034 Jump
00000010 JMP 0x00000034 Jump
00000012 JMP 0x00000034 Jump
00000014 JMP 0x00000034 Jump
00000016 JMP 0x00000045 Jump
00000018 JMP 0x00000034 Jump
0000001A JMP 0x00000034 Jump
0000001C JMP 0x00000034 Jump
0000001E JMP 0x00000034 Jump
00000020 JMP 0x00000034 Jump
00000022 JMP 0x00000034 Jump
00000024 JMP 0x00000034 Jump
00000026 JMP 0x00000034 Jump
00000028 JMP 0x00000034 Jump
--- ../../../../crt1/gcrt1.S ---
0000002A CLR R1 Clear Register
0000002B OUT 0x3F,R1 Out to I/O location
0000002C LDI R28,0x5F Load immediate
0000002D LDI R29,0x08 Load immediate
0000002E OUT 0x3E,R29 Out to I/O location
0000002F OUT 0x3D,R28 Out to I/O location
00000030 CALL 0x00000036 Call subroutine



b) Describe differences between the C and Assembly code implementations for the waveforms generated using Timers 0 and 2 (excluding version of the program is in C and the translated version is in assembly). Is the assembly code an efficient translation of the C code? Explain.

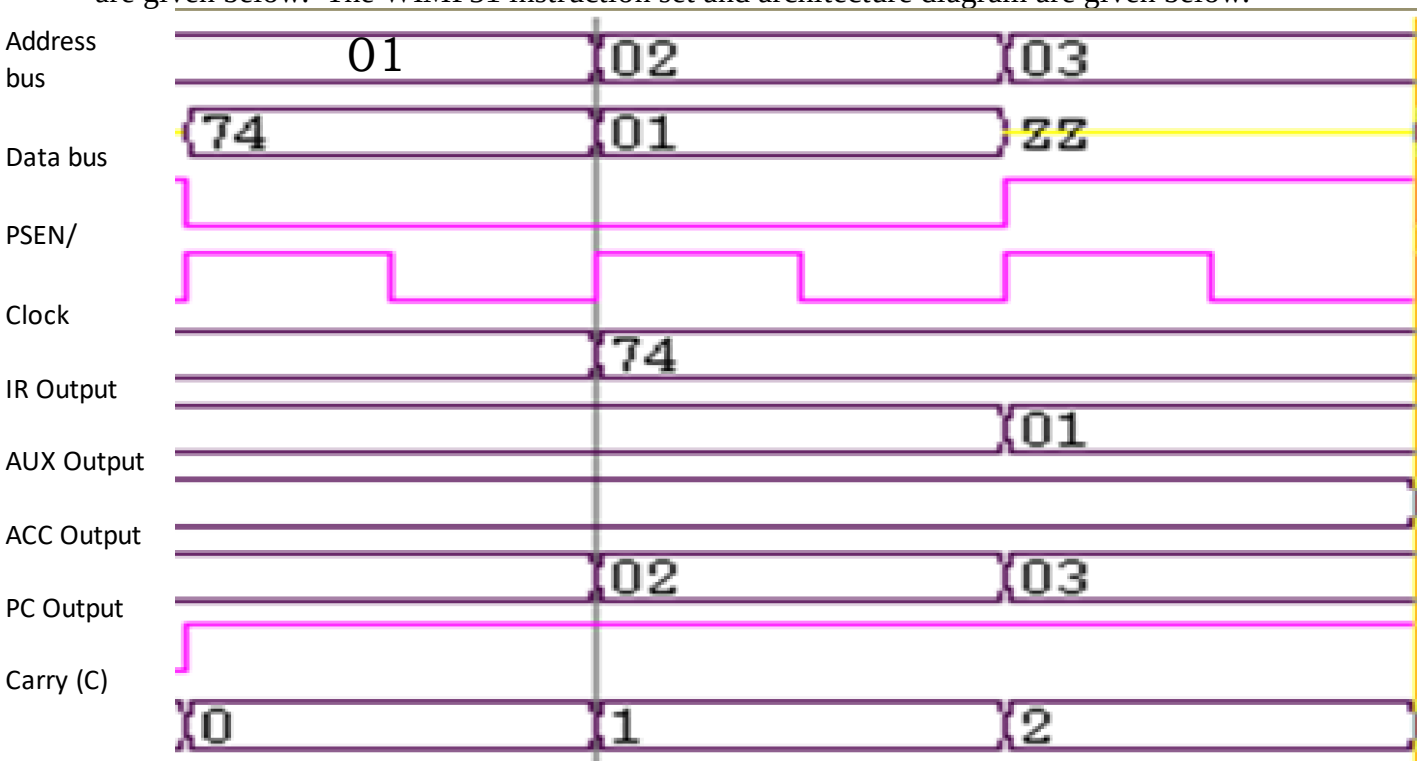
The entire vector table is given in code memory in assembly translation. This is not shown in C.

Assembly translation does not use I/O symbol labels (such as DDRB) but I/O register addresses in the instructions.

In routines including (main, timer 0 ISR, timer 2 ISR), push and pop instructions are used to maintain the status of the GPRs outside of those routines.

The assembly translation for this program is reasonable. If the program defines and manipulates variables that are not I/O registers, the code translation is less efficient. A good example of this is a multi-tiered for loop. The assembly language translation will look different than what would you likely expect.

3) Given the following timing signals for the WIMP51 for an instruction cycle. For the Current State signal, note that 0 = fetch state, 1 = decode state, 2 = execute state. The instruction set and architecture diagram are given below. The WIMP51 instruction set and architecture diagram are given below.



Determine the instruction being executed during the instruction cycle.

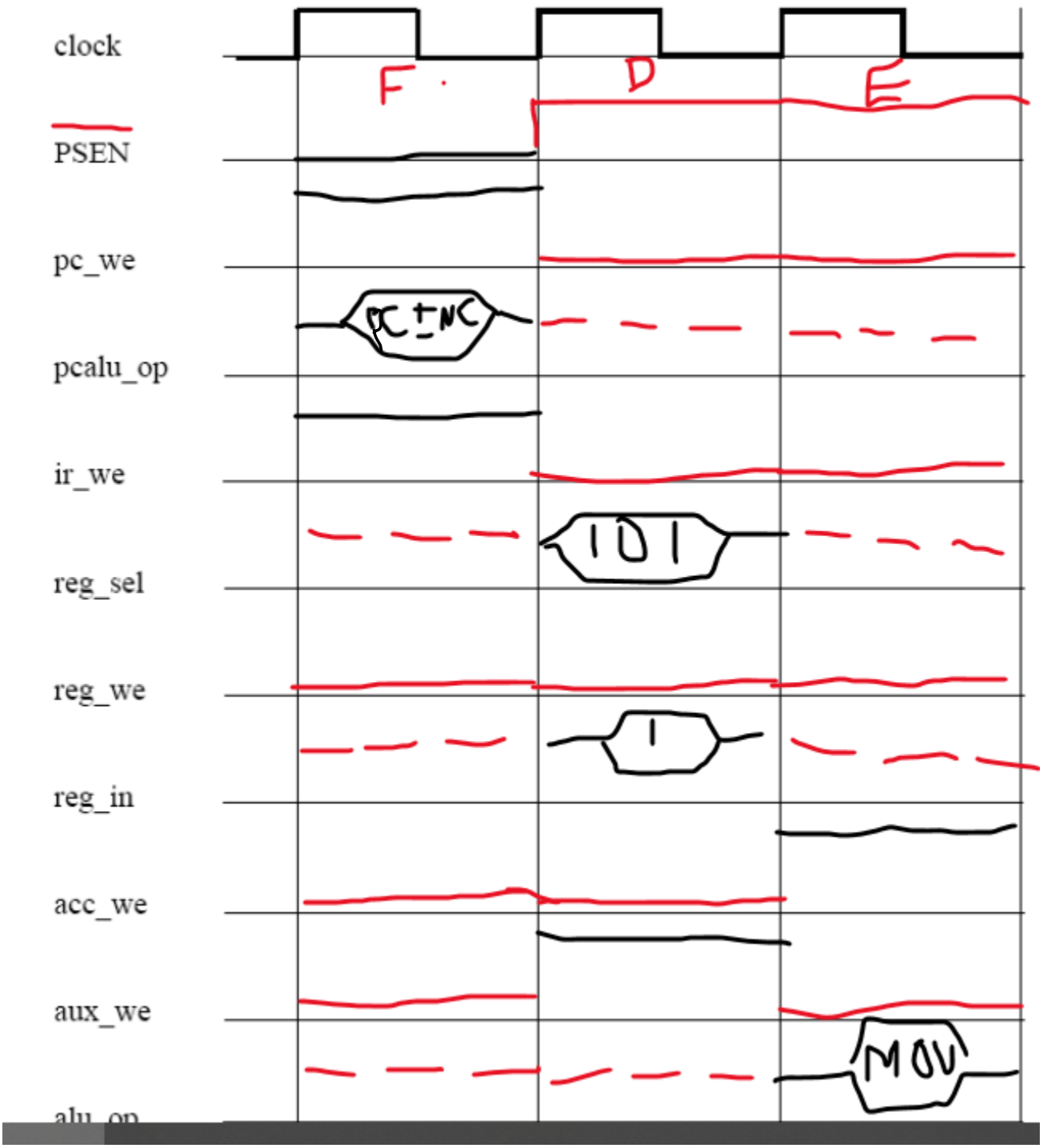
Instruction: MOV A, #01H

```
MOV A, #D   01110100 dddddddd   A<=D
```

4) Draw the timing diagram for the following control signals on the diagram below when the WIMP51 is executing the instruction ___MOV A, R5_____. Label the fetch, decode and execute parts of the instruction cycle on the clock portion of the signal below.

The dotted lines have been drawn as a graphical reference only and do not reflect the value of the corresponding signal. If you think a particular control signal is of no significance during a particular time period, indicate this opinion by using a **dashed** line to denote the signal as a "don't care" for that time period. Give "command" values for pcalu_op and alu_op; for example, indicate the value of "pcalu_op" as "PC_INC" or "PC_ADD" rather than representing this command by a numeric code. If you are not sure

about a particular control signal, make an educated guess, indicate that the value you have marked is a guess, and explain why you made this particular guess. Remember, PSEN/ pulses low whenever a byte is being fetched from code memory.

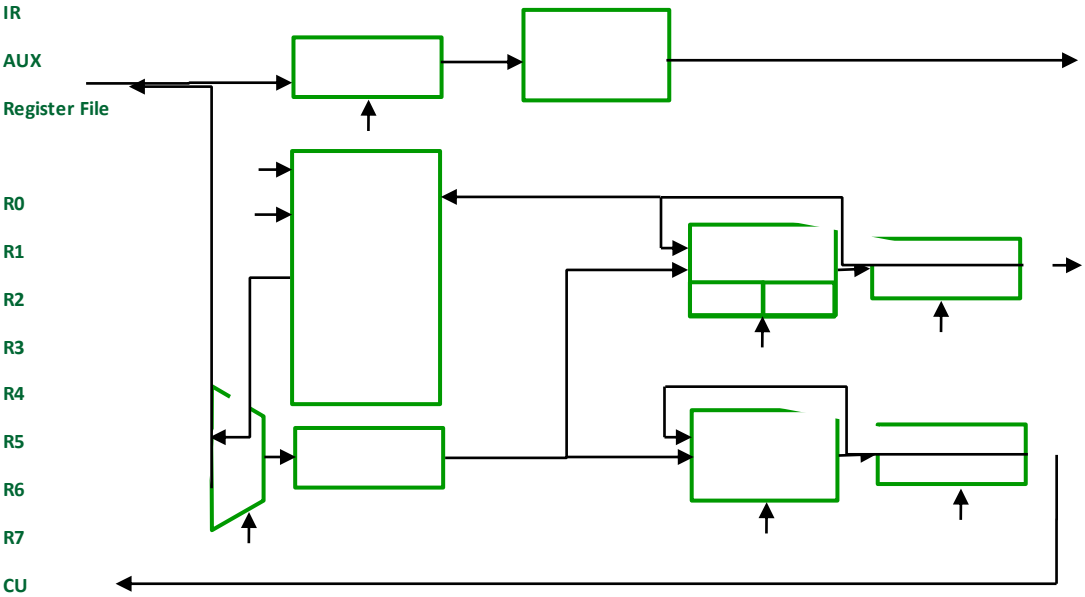


Summary of WIMP51 Inst Set

ASM code	Machine code	Meaning
MOV A, #D	01110100 dddddddd	A<=D
ADDC A,#D	00110100 dddddddd	C,A<=A+D+C

MOV Rn, A	11111nnn	Rn<=A
MOV A, Rn	11101nnn	A<=Rn
ADDC A, Rn	00111nnn	C,A<=A+Rn+C
ORL A, Rn	01001nnn	A<=A OR Rn
ANL A, Rn	01011nnn	A<= A AND Rn
XRL A, Rn	01101nnn	A<=A XOR Rn
SWAP A	11000100	$A<=A_{(3-0)}\&A_{(7-4)}$
CLR C	11000011	C<=0
SETB C	11010011	C<=1
SJMP rel	10000000 aaaaaaaa	PC<=PC+rel+2
JZ rel	01100000 aaaaaaaa	PC<=PC+rel+2 if Z=1

WIMP51 Architecture Diagram



aux_we

reg_in

ir_we

reg_we

reg_sel

reg_sel

0

1

acc_out

PSEN

5) Chapter 11: 7, 15, 30, 34 a,c, 36-polling method (no interrupts), 36-use serial interrupts (XTAL = 10 MHz)

7) FALSE

15) 4,000,000 bits (200 x 80 x 25 x 10)

30) UDR (USART Data Register), UCSRA,UCSRB, UCSRC (USART Control Status Register), and UBRR (USART Baud Rate Register)

34) X = (10 MHz/(16(Desired Baud Rate))) – 1
X = (625 kHz/(Desired Baud Rate)) – 1.
(a) (625 kHz/9600) – 1= 64 = 0x40
(c) (625 kHz/1200) – 1= 519 = 0x207

Chapter 11: 36-polling method (no interrupts), 36-use serial interrupts (XTAL = 10 MHz)

BAUD RATE = FOSC/(16(X+1))

X = FOSC/(16 BAUD RATE) – 1 = 65.1 – 1 = 64

NO INTERRUPTS-POLLING

```
.ORG 0
LDI R16,(1<<TXEN) ;enable transmitter
OUT UCSRB, R16 ;
LDI R16,(1<<UCSZ1)|(1<<UCSZ0)|(1<<URSEL) ;8-bit data
OUT UCSRC, R16 ;no parity, 1 stop bit
LDI R16,64 ;9600 baud rate (X = 64)
OUT UBRRL,R16 ;for XTAL = 10 MHz
AGAIN:
      SBIS UCSRA,UDRE ;is UDR empty
      RJMP AGAIN ;wait more
      LDI R16,'G' ;send 'G'
      OUT UDR,R16 ;to UDR
      RJMP AGAIN ;do it again
```

SERIAL INTERRUPTS

```
.ORG 0
RJMP MAIN
.ORG 0x1C
RJMP USART0_UDRE

.ORG 0x100
MAIN:       LDI R16,(1<<UCSZ1)|(1<<UCSZ0)|(1<<URSEL) ;8-bit data
      OUT UCSRC, R16 ;no parity, 1 stop bit
      LDI R16,64 ;9600 baud rate (X = 64)
      OUT UBRRL,R16 ;for XTAL = 10 MHz
      LDI R16,(1<<UDRIE) | (1<<TXEN);  enable serial transmission and interrupt for when value to
                                    ; be transmitted in UDR is shifted out (UDR is empty)
                                    ; UCSZ2 = 0 for setting the number of bits to 8
      OUT UCSRB, R16;
      SEI; I=1 GLOBAL INTERRUPT ENABLE
      HERE: RJMP HERE; endless loop
.ORG 0x150
USART0_UDRE: LDI R17, 'G'
      OUT UDR, R17; send ASCII 'G' to UDR
      RETI
```

Problems are below.

SECTION 11.1: BASICS OF SERIAL COMMUNICATION

1. Which is more expensive, parallel or serial data transfer?
2. True or false. 0- and 5-V digital pulses can be transferred on the telephone without being converted (modulated).
3. Show the framing of the letter ASCII 'Z' (0101 1010), no parity, 1 stop bit.
4. If there is no data transfer and the line is high, it is called _____ (mark, space).
5. True or false. The stop bit can be 1, 2, or none at all.
6. Calculate the overhead percentage if the data size is 7, 1 stop bit, and no parity bit.
7. True or false. The RS232 voltage specification is TTL compatible.
8. What is the function of the MAX 232 chip?
9. True or false. DB-25 and DB-9 are pin compatible for the first 9 pins.
10. How many pins of the RS232 are used by the IBM serial cable, and why?
11. True or false. The longer the cable, the higher the data transfer baud rate.
12. State the absolute minimum number of signals needed to transfer data between two PCs connected serially. What are those signals?
13. If two PCs are connected through the RS232 without a modem, both are configured as a _____ (DTE, DCE) -to- _____ (DTE, DCE) connection.

14. State the nine most important signals of the RS232.
15. Calculate the total number of bits transferred if 200 pages of ASCII data are sent using asynchronous serial data transfer. Assume a data size of 8 bits, 1 stop bit, and no parity. Assume each page has 80 × 25 of text characters.
16. In Problem 15, how long will the data transfer take if the baud rate is 9600?

29. Which of the following baud rates are supported by the HyperTerminal program in PC?
    (a) 4800 (b) 3600 (c) 9600
    (d) 1800 (e) 1200 (f) 19,200
30. Which register of ATmega32 is used for baud rate programming?
31. Which bit of the UCSRA is used for baud rate speed?
32. What is the role of the UDR register in serial data transfer?
33. UDR is a(n) ____-bit register.
34. For XTAL = 10 MHz, find the UBRR value (in both decimal and hex) for each of the following baud rates.
    (a) 9600 (b) 4800 (c) 1200
35. What is the baud rate if we use UBRR = 15 to program the baud rate? Assume XTAL = 10 MHz.
36. Write an AVR program to transfer serially the letter 'Z' continuously at 9600 baud rate. Assume XTAL = 10 MHz.
37. When is the PE flag bit raised?
38. When is the RXC flag bit raised or cleared?
39. When is the UDRE flag bit raised or cleared?
40. To which register do RXC and UDRE belong?
41. Find the UBRR for the following baud rates if XTAL = 16 MHz and U2X = 0.

    (a) 9600 (b) 19200
    (c) 38400 (d) 57600
42. Find the UBRR for the following baud rates if XTAL = 16 MHz and U2X = 1.
    (a) 9600 (b) 19200
    (c) 38400 (d) 57600

SECTION 11.4: AVR SERIAL PORT PROGRAMMING IN C

49. Write an AVR C program to transmit serially the letter 'Z' continuously at 9600 baud rate.
50. Write an AVR C program to transmit serially the message "The earth is but one country and mankind its citizens" continuously at 57,600 baud rate.